

Nonabelian Completion Holonomy and Sharp LIFO Carrier Costs: A Free–Toral Bifurcation

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Abstract

We develop a category-relative bridge from optimal-completion topology to physical LIFO memory. For a completable residual with active lattice L , the filled shortest-prefix quotient carries a canonical displacement homomorphism $\delta : \pi_1(I) \rightarrow L$. We isolate the zero-displacement loop group $K_0 = \ker \delta$ and prove an exact active-rank-two visibility trichotomy: the toral branch has $K_0 = 1$; in the free F_2 branch the pure sector is the commutator subgroup and is invisible to first homology; in the free F_3 branch one pure direction is already visible in H_1 .

We then study a four-letter scalar completion system with two residuals having the same equal-length kernel, active lattice, rank, and minimum completion length but different optimal-fiber topology. At the free residual, a Stable-LIFO comparison transport is locally flat and sends a zero-displacement commutator to a nontrivial 3-cycle. We realize this holonomy on the full shortest-completion episode by actual literal stack order, without epsilon moves or a permanent sidecar. In the literal-stack-only persistence category, every nonempty zero-weight order carrier has height at least d , and an explicit construction attains this bound. The chosen realization uses two transient ancestry handles and three deferred path states.

At the toral residual, the same raw Stable-LIFO principle is locally curved. Full torus descent forces commuting generator images and yields an exact global correction Pareto law: the nondominated profiles are $(3, 6, 4)$ and $(4, 4, 2)$ for altered endpoint comparisons, total transposition distance, and maximum correction support. The nearest correction compresses to a cyclic C_3 image and uses the same height- d carrier, while complete erasure of transport is possible at correction distance 8. The free and toral branches therefore control where noncommutativity can reside: as global pure holonomy when the commutator cell is absent, or as correction cost when it is filled.

Keywords. deterministic pushdown automata; shortest completion; optimal fibers; fundamental groups; holonomy; LIFO memory; resource lower bounds.

1 Introduction

The completion geometry of a pushdown normalizer can contain more information than the scalar deficit that determines how many symbols remain to be supplied. In higher-dimensional minimum-completion fibers, equal-length trade directions produce loops. The central question of this paper is whether such loops can be given an operational meaning in LIFO memory.

The standard algebraic-topology and free-group facts used below are recalled from [1, 2], while the pushdown-machine background is standard [3]. A naive statement would be false: nontrivial topology does not force physical stack debt in an unrestricted machine, because finite control or an unrelated re-coordination may trivialize a chosen transport. The correct question is category-relative. Once a physical comparison policy is fixed, which zero-displacement loops

can carry nontrivial order holonomy, and what literal-stack resources are required to realize that holonomy?

We answer this in active rank two. For the three rank-two topological alternatives that arise in the optimal-fiber classification, the structural theorem computes the kernel of the canonical displacement map. The fixed free/toral family used for the operational results is verified explicitly in this paper, so no edgewise physical interpretation is inferred merely from a homotopy compression. Cubical and higher-dimensional automata provide a broader topological language for commuting executions [4, 5]; our connection is instead built on shortest-completion fibers and then realized by literal stack order. We then give a single explicit completion system in which two residuals exhibit opposite operational behavior: the free residual supports pure nonabelian LIFO holonomy, while the toral residual turns the same raw noncommutativity into a correction problem.

2 From the original 2:1 machine to completion holonomy

The topological question studied here arose from a concrete pushdown computation rather than from an abstract cell complex. The author's original completion-aware 2:1 DPDA is shown in fig. 1. Its center/left/right states and stack jointly record what remains to be supplied to restore the target balance, and its end-of-input rules rematerialize the finite correction carried by a side state as explicit stack demand.

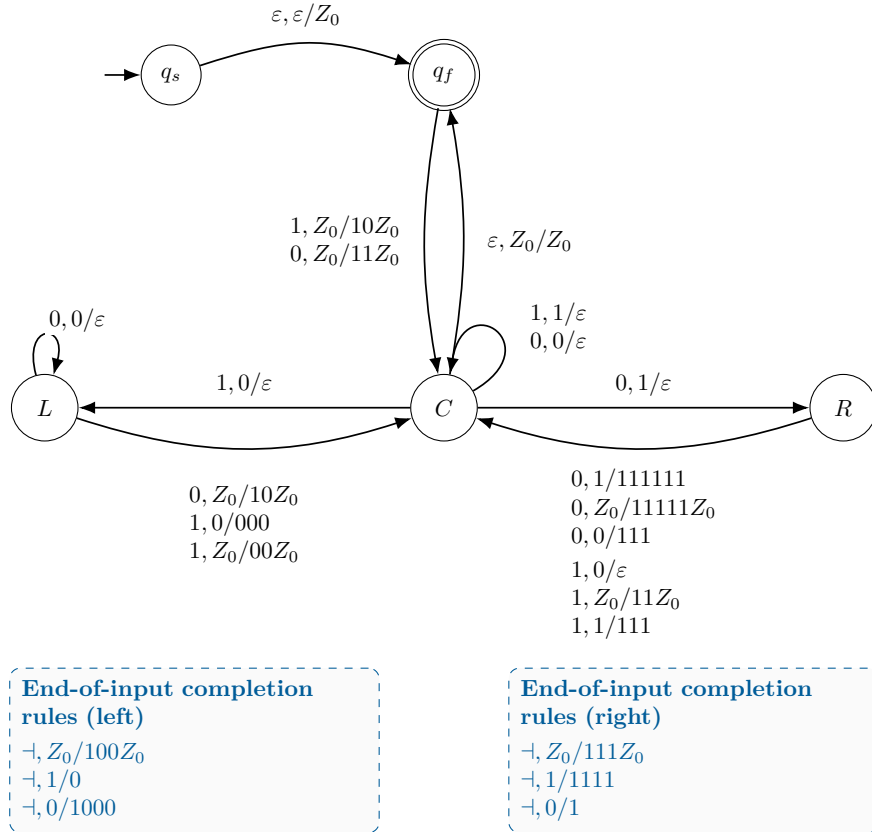


Figure 1: The original completion-aware 2:1 DPDA, in the empty-word-included formal variant. Its center/left/right structure makes completion demand a joint control-and-stack quantity. The present topology-to-LIFO theory is reached by abstracting this completion semantics and then passing to higher-dimensional optimal-completion fibers.

The $p:q$ generalization turns this mechanism into an exact residual and shortest-completion

semantics. In the binary rational family the minimum demand is essentially one-dimensional, but the same completion viewpoint applied to several weighted letters produces minimum-completion *fibers*: distinct optimal count vectors can represent completions of the same length, and equal-length kernel trades generate loops in the filled shortest-prefix quotient. This is the point at which a question that was invisible in the original 2:1 table appears: can a zero-displacement loop be carried physically as nontrivial LIFO order memory?

Accordingly, the lineage of the present paper is

$$\boxed{\begin{array}{l} 2:1 \text{ completion-aware stack} \longrightarrow p:q \text{ residual/shortest-completion semantics} \\ \longrightarrow \text{higher-dimensional optimal fibers} \longrightarrow \text{holonomy and LIFO carrier cost} \end{array}}$$

The original automaton is therefore motivation and provenance, not a proof premise. The theorems below start from the optimal-fiber/displacement formalism and then return to an explicit physical stack realization.

3 Optimal fibers, displacement, and pure loops

Let $\mathcal{O}(r) \subset \mathbb{N}^s$ be the set of minimum-length completion count vectors at residual r . Fix $x_0 \in \mathcal{O}(r)$ and put

$$P = \mathcal{O}(r) - x_0, \quad L = \langle \mathcal{O}(r) - \mathcal{O}(r) \rangle_{\mathbb{Z}}.$$

Let $I = I(r)$ denote the filled shortest-prefix quotient associated with this optimal fiber. In active rank two, the completion-fiber construction gives a canonical surjection

$$\delta : \pi_1(I) \twoheadrightarrow L \cong \mathbb{Z}^2,$$

which records the lattice displacement of a loop.

Definition 3.1 (Pure-loop sector). *Define*

$$K_0 = \ker \delta$$

and

$$K_{\text{lin}} = \ker (H_1(I; \mathbb{Z}) \rightarrow L).$$

The first group is the full zero-displacement sector; the second records the part already visible after abelianization.

Theorem 3.2 (Rank-two displacement-kernel visibility trichotomy). *Assume rank $L = 2$. Then the active-rank-two alternatives have the following pure-loop structure.*

1. *If I is toral, $\pi_1(I) \cong \mathbb{Z}^2$ and δ is an isomorphism. Hence*

$$K_0 = 1, \quad K_{\text{lin}} = 0.$$

2. *If $I \simeq S^1 \vee S^1$, then $\pi_1(I) \cong F_2$ and*

$$K_0 = [F_2, F_2], \quad K_{\text{lin}} = 0.$$

Thus pure topology exists but is nonabelian-only.

3. *If $I \simeq S^1 \vee S^1 \vee S^1$, then $\pi_1(I) \cong F_3$ and*

$$K_{\text{lin}} \cong \mathbb{Z}.$$

The full kernel $K_0 = \ker(F_3 \twoheadrightarrow \mathbb{Z}^2)$ is free of countably infinite rank, but one pure direction is already visible in H_1 .

Proof. In the toral branch the displacement theorem identifies $\pi_1(I)$ with L . In the F_2 branch, abelianization gives a surjection $\mathbb{Z}^2 \rightarrow \mathbb{Z}^2$, hence an isomorphism, so the kernel of $F_2 \rightarrow \mathbb{Z}^2$ is exactly the commutator subgroup. In the F_3 branch, abelianization gives a surjection $\mathbb{Z}^3 \rightarrow \mathbb{Z}^2$ with primitive rank-one kernel. The assertions about the full kernels follow from Nielsen–Schreier and the standard structure of infinite-index normal subgroups of nonabelian free groups. $\square$

Definition 3.3 (Displacement-compatible physical transport). *A flat physical transport is a functor from the fundamental groupoid $\Pi_1(I)$ to a groupoid of invertible physical comparison maps. At a basepoint it induces*

$$\text{Hol} : \pi_1(I) \rightarrow G_{\text{phys}}.$$

It is displacement-compatible if there is a homomorphism $c : G_{\text{phys}} \rightarrow L$ such that

$$c \circ \text{Hol} = \delta.$$

Then $\text{Hol}(K_0)$ lies in the count-invisible subgroup $\ker c$.

Proposition 3.4 (Gauge persistence of pure holonomy). *If $\text{Hol}|_{K_0}$ is nontrivial, no objectwise change of physical trivialization within the same flat invertible transport category can make every pure loop trivial. Such a change only conjugates the holonomy representation.*

This proposition is deliberately category-relative. It does not say that topology forces debt in an arbitrary DPDA.

4 A fixed completion system with a free/toral residual bifurcation

Fix $d \geq 5$ and the four weights

$$\omega_d = (1 - 2d, 1 - d, 1, 1 + d).$$

The equal-length kernel contains the basis

$$u = (1, -2, 1, 0), \quad v = (2, -3, 0, 1).$$

Consider the two residuals

$$r_W = 5d - 4, \quad r_T = 3d - 4.$$

Both have minimum completion length 4 and the same active rank two.

At r_W the optimal completions are

$$x_0 = (1, 3, 0, 0), \quad x_u = (2, 1, 1, 0), \quad x_v = (3, 0, 0, 1),$$

so the normalized fiber is $\{0, u, v\}$ and

$$I(r_W) \simeq S^1 \vee S^1.$$

At r_T the optimal completions are

$$(0, 3, 1, 0), \quad (1, 1, 2, 0), \quad (1, 2, 0, 1), \quad (2, 0, 1, 1),$$

which form a lattice parallelogram; hence

$$I(r_T) \simeq T^2.$$

By Theorem 3.2, the first residual has a nonabelian-only pure sector $[F_2, F_2]$, whereas the second has no pure sector at all.

5 Stable-LIFO comparison transport

For an equal-length count vector $x = (x_1, \dots, x_s)$ define the slot word

$$\eta_{\text{slot}}(x) = 1^{x_1} 2^{x_2} \dots s^{x_s}.$$

Temporarily label its slots from left to right. For two equal-length count vectors x, y , define the Stable-LIFO matching $\tau_{x,y}$ by preserving same-type occurrences stably from left to right and matching all remaining source positions to remaining target positions in reverse order. Then

$$\tau_{y,x} = \tau_{x,y}^{-1}.$$

For the free optimal endpoints, the canonical words are

$$w_0 = 1222, \quad w_u = 1123, \quad w_v = 1114.$$

The three comparison permutations are

$$\tau_{0u} = (234), \quad \tau_{0v} = (24), \quad \tau_{uv} = (34).$$

On moved slots $\{2, 3, 4\}$ identify

$$r = (012), \quad s = (02).$$

Then

$$[r, s] = (021),$$

corresponding on the original slots to (243).

Theorem 5.1 (Pure nonabelian Stable-LIFO holonomy). *The Stable-LIFO comparison transport on the free optimal fiber is flat and sends a zero-displacement commutator in $K_0 = [F_2, F_2]$ to the nontrivial 3-cycle*

$$(243).$$

The image is invisible to every abelian permutation observable.

Proof. The triangle relation is

$$\tau_{0u}\tau_{uv} = \tau_{0v}$$

under chronological right action, so the endpoint comparisons define a flat local system on the three-point free quotient. The two displacement generators map to r and s , and their commutator maps to $(021) \neq 1$. Since every homomorphism from S_4 to an abelian group kills commutators, the certificate is genuinely nonabelian. $\square$

6 The full free shortest-completion episode

A prefix count vector y determines

$$n = \mathbf{1}^\top y, \quad s = y_2 + 2y_3 + 3y_4.$$

For r_W ,

$$r_W + \omega_d \cdot y = r_W + (1 - 2d)n + ds.$$

Since $0 \leq n \leq 4 < d$, residual equality on shortest prefixes is equivalent to equality of (n, s) . The full quotient therefore has vertex set

$$V_W = \{(0, 0)\} \cup \{(1, s) : 0 \leq s \leq 3\} \cup \{(2, s) : 0 \leq s \leq 3\} \cup \{(3, s) : 0 \leq s \leq 3\} \cup \{(4, 3)\},$$

so $|V_W| = 14$. Its cell vector is

$$(f_0, f_1, f_2, f_3) = (14, 28, 15, 2).$$

Theorem 6.1 (Full free-episode realization). *The entire 14-vertex shortest-prefix quotient admits a flat S_3 connection whose endpoint products are*

$$w_0 \mapsto e, \quad w_u \mapsto r, \quad w_v \mapsto s.$$

All square relations and both 3-cube boundaries are flat. The pure commutator of Theorem 5.1 therefore survives on the actual one-letter shortest-completion episode, not merely on a compressed comparison complex.

Construction. Assign edge labels by a spanning-tree gauge and extend them over the remaining edges so that each of the 15 squares has trivial boundary product. The two 3-cubes then have flat boundary automatically. The explicit connection uses only the four local values e, r, r^{-1}, s and the transposition $a = (01)$ at the mixed vertices; direct multiplication gives the endpoint products above for all 20 shortest words. $\square$

7 Sharp literal-stack carrier cost

Before measuring a carrier, we fix the implementation category. A *buffered ancestry refinement* of capacity b allows at most b temporary finite-control handles that remember distinguished stack-cell ancestries during a one-top macro; the handles are cleared at the endpoint. A macro is ancestry-bijective if every initial distinguished ancestry has exactly one final descendant. Appendix B proves the general bottom-move lower bound $b \geq n - 1$ when the bottom of an n -cell distinguished prefix changes ancestry. We now impose a specific physical persistence category: no epsilon moves, no permanent finite-control sheet register, common endpoint control and zero potential, literal stack alphabet $\{1, 2, 3, 4\}$ only, and identical endpoint Parikh counts across sheets. Persistent monodromy must therefore live in stack order.

For a zero-weight literal count vector $z \in \mathbb{N}^4$, define its order capacity by the multinomial number

$$M(z) = \frac{|z|_1!}{\prod_i z_i!}.$$

For $k \geq 2$, set

$$g_k^{\text{ord}}(\omega_d) = \min\{|z|_1 : z \neq 0, \omega_d z = 0, M(z) \geq k\}.$$

Lemma 7.1 (Arithmetic carrier girth). *Every nonempty zero-weight literal word has length at least d . Moreover*

$$g_3^{\text{ord}}(\omega_d) = g_6^{\text{ord}}(\omega_d) = d.$$

Proof. Each weight is congruent to 1 modulo d . Hence $\omega_d z = 0$ implies

$$|z|_1 \equiv 0 \pmod{d},$$

so every nonempty zero-weight word has length at least d . Equality is attained by

$$z^{(3)} = (0, 1, d-1, 0)$$

for three orderings and by

$$z^{(6)} = (0, 2, d-3, 1)$$

for six orderings. $\square$

For the six-state carrier, let

$$T_d = 2 \cdot 3^{d-4}$$

and index the six permutations P_g of the three letters 234 by $g \in S_3$. Set

$$Z_g = P_g T_d.$$

All Z_g have the same Parikh vector $z^{(6)}$ and zero total weight.

Theorem 7.2 (Sharp free literal carrier). *There is a real-time one-top realization of the full free connection such that*

$$Z_g \xrightarrow{w_0} Z_g, \quad Z_g \xrightarrow{w_u} Z_{gr}, \quad Z_g \xrightarrow{w_v} Z_{gs},$$

with no epsilon moves and no permanent sidecar. The minimum persistent literal order-carrier height in the stated category is exactly

$$\boxed{d}.$$

The chosen realization needs exactly two transient ancestry handles. In the deferred-carrier subclass, exactly three transient path states are necessary and sufficient.

Proof. The lower bound on persistent height is Lemma 7.1: every nonempty zero-weight literal carrier has length divisible by d , hence length at least d . For the upper bound use the six same-Parikh words $Z_g = P_g T_d$, $g \in S_3$. Only the distinguished three-cell prefix P_g is changed; the common suffix T_d remains untouched throughout.

For a shortest word w , let $\sigma(w)$ be the endpoint product supplied by the flat connection of Theorem 6.1. During the four input-synchronous steps, finite control records the current quotient vertex and two temporary ancestry handles. The first two distinguished carrier cells are exposed into those handles, the third remains visible, and on the terminal step the visible third cell together with the two handles determines the incoming prefix permutation g . The transition rewrites that visible cell by the three-cell prefix $P_{g\sigma(w)}$. Thus

$$Z_g \xrightarrow{w_0} Z_g, \quad Z_g \xrightarrow{w_u} Z_{gr}, \quad Z_g \xrightarrow{w_v} Z_{gs},$$

with one input symbol consumed per transition and no ε -move. The semantic shortest-episode coordinate in control supplies the unique potential needed to verify the residual identity at each step, so this is an exact lift. Appendix A gives the full edge certificate for all twenty shortest words.

Necessity of two ancestry handles follows from Lemma B.1: the generator $s = (02)$ changes the ancestry occupying the bottom of the distinguished three-cell prefix, so any ancestry-bijjective top-access macro needs at least $3 - 1 = 2$ handles. The construction above uses two.

Finally restrict to the deferred-carrier subclass, in which the distinguished carrier order is not changed before the terminal rewrite. The three histories 222, 231, and 411 reach the same preterminal semantic vertex with the same next input but with accumulated products s, a, e ; their common last input must therefore produce three distinct endpoint multipliers e, r, s . Determinism forces three transient path states. The explicit connection table has at most three accumulated-product classes at every vertex, so three states suffice. Hence both locality bounds are sharp in the stated subclasses. $\square$

8 Toral curvature and the difference between local flattening and torus descent

For the toral residual write the four optimal endpoints as $0, g, h, g + h$. Raw Stable-LIFO endpoint comparisons are

$$A = (1243), \quad B = (24), \quad C = (1234), \quad D = (23).$$

The two chronological route products are

$$AB = (143), \quad CD = (134) = (143)^{-1}.$$

Thus the raw parallelogram has nontrivial 3-cycle curvature.

On one isolated square, one edge change is enough to make the two route products equal. But a torus local system must descend globally. If normalized path products to the three nonzero lattice points are U, V, W , torus descent requires

$$[U, V] = e, \quad W = UV = VU.$$

Hence the corrected endpoint tuple must have the form

$$(\tilde{A}, \tilde{B}, \tilde{C}, \tilde{D}) = (U, V, V, U), \quad UV = VU.$$

Theorem 8.1 (Minimum number of globally altered endpoint comparisons). *Any torus-descending correction of the raw tuple (A, B, C, D) changes at least three of the four endpoint comparisons. Three changes suffice.*

Proof. Because $A \neq D$, at least one of the g -pair must change; because $B \neq C$, at least one of the h -pair must change. If exactly two comparisons changed, the retained commuting pair would have to be one of $(A, B), (A, C), (D, B), (D, C)$. Each pair has nontrivial commutator (243), so two changes are impossible. Taking $U = A$ and $V = e$ gives (A, e, e, A) and changes exactly three entries. $\square$

Let d_T be Cayley distance in S_4 with respect to transpositions and define

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V)$$

subject to $[U, V] = e$.

Theorem 8.2 (Exact global toral correction Pareto law). *The minimum total correction distance is*

$$\boxed{F_{\min} = 4},$$

attained by

$$U = (243), \quad V = (234) = U^{-1},$$

with four single-transposition corrections. Among corrections changing exactly three endpoint comparisons, the minimum total transposition length is 6 and the minimum possible maximum support is 4. Hence the exact nondominated profiles

$$(\# \text{ altered}, \text{ total transposition length}, \max \text{ support})$$

are

$$\boxed{(3, 6, 4), \quad (4, 4, 2)}.$$

Proof. For any commuting U, V ,

$$F(U, V) \geq d_T(A, D) + d_T(B, C) = 2 + 2 = 4$$

by the triangle inequality. Equality is realized by the displayed U, V , using

$$A(12) = U, \quad B(34) = V, \quad C(12) = V, \quad D(34) = U.$$

For the three-change branch, retain one raw edge and minimize over its centralizer. The four symmetric cases give minimum total length 6 and force one correction to move all four slots. These two profiles are incomparable and every other correction is dominated by one of them. $\square$

9 The full toral episode and repair/debt tradeoff

For r_T the residual quotient again uses (n, s) , now with vertex set

$$V_T = \{(0, 0)\} \cup \{(1, s) : 0 \leq s \leq 3\} \cup \{(2, s) : 0 \leq s \leq 5\} \cup \{(3, s) : 2 \leq s \leq 5\} \cup \{(4, 5)\}.$$

Thus

$$|V_T| = 16, \quad (f_0, f_1, f_2, f_3) = (16, 38, 28, 6).$$

There are 40 shortest completion words.

For the minimum-distance correction of Theorem 8.2, $V = U^{-1}$ and the image is the cyclic group

$$\langle U \rangle \cong C_3.$$

Define on the active lattice

$$\rho_T(ag + bh) = U^{a-b}.$$

Because the toral displacement map is an isomorphism, this defines a flat local system on the entire quotient. The 40 shortest words split into endpoint products

$$16 \text{ words} \mapsto e, \quad 12 \mapsto U, \quad 12 \mapsto V.$$

Theorem 9.1 (Full toral realization and exact repair/debt tradeoff). *The minimum-distance toral repair admits a full 16-vertex real-time realization with nontrivial displacement-visible C_3 transport. If this transport is stored persistently in literal stack order, the minimum carrier height is d ; the three-slot carrier uses two ancestry handles and, in the deferred subclass, three transient path states.*

Complete erasure of transport is possible by setting $U = V = e$. Its total correction distance from the raw tuple is

$$\boxed{8}.$$

Thus toral topology forces no pure debt: one may retain the nearest nontrivial flat transport at distance 4, or pay distance 8 and remove persistent order memory entirely.

Proof. The corrected image C_3 acts only on three slots, so the free height- d carrier construction applies. The lower bound is again the zero-weight congruence. For complete erasure,

$$\ell_T(A) + \ell_T(B) + \ell_T(C) + \ell_T(D) = 3 + 1 + 3 + 1 = 8.$$

Since $K_0 = 1$ in the toral branch, none of this retained transport is pure zero-displacement holonomy; it factors through the displacement lattice. $\square$

10 The free–toral topology-to-LIFO dichotomy

The fixed family gives a clean contrast with every algebraic parameter except the residual held fixed.

free: local flatness permits global pure nonabelian holonomy,
 toral: the filled commutator cell forbids pure holonomy
 and converts raw noncommutativity into correction cost.

The distinction is not simply “topology requires memory.” In the free branch, the missing parallelogram leaves a nonabelian zero-displacement sector in which order memory can live. In the toral branch, the parallelogram fills the commutator relation, so a flat physical representation must commute on the lattice generators. Noncommutativity survives only as the cost of repairing a chosen raw comparison policy.

11 Scope and representation dependence

The following distinctions are essential.

- The group K_0 , the existence of nontrivial pure holonomy, the free/toral split, and the literal zero-weight height lower bound are intrinsic within the stated completion-fiber and carrier categories.
- The concrete permutations r, s, A, B, C, D and the correction profiles of Theorem 8.2 are relative to the fixed Stable-LIFO endpoint-comparison policy, although they are invariant under simultaneous conjugation of slot labels.
- The two-handle and three-state bounds are exact for the chosen three-slot and deferred-carrier subclasses; they are not claimed as universal lower bounds over all possible pushdown policies.
- No theorem here says that nontrivial topology alone forces physical debt under unrestricted finite control.

12 Conclusion

The displacement kernel gives the correct topological location of count-invisible completion loops. In rank two it separates toral fibers, where no pure sector exists, from free fibers, where the pure sector can be either nonabelian-only or already visible in first homology. The fixed-system example then turns that structural distinction into an operational one: a free residual supports sharp literal LIFO holonomy, whereas the toral residual requires a commuting repair and exhibits an exact correction/debt tradeoff.

A Full free-episode connection certificate

The full shortest-prefix quotient at r_W has vertices

$$V_W = \{(0, 0)\} \cup \{(1, s), (2, s), (3, s) : 0 \leq s \leq 3\} \cup \{(4, 3)\}$$

and f -vector $(14, 28, 15, 2)$. A directed edge is written $(n, s) \xrightarrow{i} (n+1, s+i-1)$. With chronological right multiplication, put

$$r = (012), \quad r^{-1} = (021), \quad s = (02), \quad a = (01).$$

The following table is an explicit flat S_3 connection; omitted edges do not exist.

source	i	σ
(0, 0)	1, 2, 3, 4	e
(1, 0)	1, 2, 3, 4	e
(1, 1)	1	e
(1, 1)	2	r^{-1}
(1, 1)	3	a
(1, 2)	1	e
(1, 2)	2	a
(1, 3)	1	e
(2, 0)	1	e
(2, 0)	2	e
(2, 0)	3	r
(2, 0)	4	e
(2, 1)	1	e
(2, 1)	2	e
(2, 1)	3	a
(2, 2)	1	r
(2, 2)	2	a
(2, 3)	1	e
(3, 0)	4	s
(3, 1)	3	r
(3, 2)	2	e
(3, 3)	1	s

Substitution into the fifteen square relations

$$\sigma_{v,i}\sigma_{v+e_i,j} = \sigma_{v,j}\sigma_{v+e_j,i}$$

uses only $r^{-1}r = e$ and $as = r$ besides identities with the same factor on both sides. Flatness of all square faces implies compatibility on the two 3-cube boundaries. Endpoint products are e, r, s on the three optimal boxes, so the commutator image is $(021) \neq e$.

B Ancestry model and sharp local-memory bounds

A buffered ancestry refinement of capacity b augments finite control with at most b temporary handles. A one-top microstep may move the visible cell's ancestry into a handle, attach a stored handle to the rewritten visible cell, or push a new decorated cell carrying a stored handle. A closed macro is ancestry-bijective when every initial distinguished ancestry has exactly one final descendant.

Lemma B.1 (Bottom-move bound). *If an ancestry-bijective top-access macro on n distinguished cells changes the ancestry occupying the bottom distinguished position, then $b \geq n - 1$. The bound is sharp.*

Proof. Before the original bottom cell can be inspected, all $n - 1$ cells above it have been removed from the live stack. Their ancestry classes must remain available for the final bijection, hence all $n - 1$ must be retained externally at that instant. Conversely, pop the upper $n - 1$ ancestries into handles, rewrite the exposed bottom, and rebuild the prefix in the target order. $\square$

For the free three-slot carrier, the generator $s = (02)$ moves the third distinguished cell, so two handles are necessary and sufficient. The strict-streaming no-go on the full interval follows

because, before the third ancestry is visible, every second-layer carrier update lies in a common order-two stabilizer. Flatness then forces the two endpoint generators into a common conjugate of that stabilizer, hence forces them to commute. The noncommuting pair r, s therefore requires transient path memory in addition to the two ancestry handles.

In the deferred-carrier subclass, the three histories 222, 231, and 411 reach the common preterminal vertex with accumulated products s, a, e and share the same final input. They must be distinguished before the final rewrite, giving a lower bound of three transient path states. The edge table above has local accumulated-product sets of size at most three, so three states suffice.

C Global torus-descent audit

For the toral residual, a normalized flat endpoint quadruple has the form

$$(U, V, V, U), \quad UV = VU.$$

Relative to the raw tuple

$$A = (1243), \quad B = (24), \quad C = (1234), \quad D = (23),$$

at least three entries must change: $A \neq D$, $B \neq C$, and every possible retained pair from $\{A, D\} \times \{B, C\}$ has nontrivial commutator. Three changes are attained by (A, e, e, A) .

For Cayley distance d_T with all transpositions as generators,

$$F(U, V) = d_T(A, U) + d_T(D, U) + d_T(B, V) + d_T(C, V) \geq 4.$$

Equality is attained by

$$U = (243), \quad V = (234) = U^{-1},$$

through

$$A(12) = U, \quad B(34) = V, \quad C(12) = V, \quad D(34) = U.$$

If exactly three raw comparisons are altered, one raw edge is retained. The four centralizer cases give:

retained	fixed generator	min F	min max supp
A	$U = A$	6	4
D	$U = D$	6	4
B	$V = B$	6	4
C	$V = C$	6	4

so the nondominated profiles are $(3, 6, 4)$ and $(4, 4, 2)$. The minimum-distance pair generates C_3 and acts on only three distinguished slots; therefore it reuses the exact height- d three-slot carrier with two ancestry handles. The full toral quotient has vertex set

$$\{(0, 0)\} \cup \{(1, s) : 0 \leq s \leq 3\} \cup \{(2, s) : 0 \leq s \leq 5\} \cup \{(3, s) : 2 \leq s \leq 5\} \cup \{(4, 5)\}$$

and f -vector $(16, 38, 28, 6)$. The representation $\rho_T(ag + bh) = U^{a-b}$ therefore gives a flat connection on all 40 shortest words.

Statements and Declarations

Author contributions. Alp Eren Bütün is the sole author and was responsible for conceptualization, methodology, formal analysis, investigation, theorem development, computational validation where reported, visualization, writing—original draft, and writing—review and editing.

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Data and code availability. The paper is theoretical and uses no empirical dataset. All definitions, constructions, formulas, finite certificates, and proofs required for the mathematical results are contained in the manuscript. Any computations reported in the manuscript are independent regression checks and are not proof premises. No external dataset or executable code is required to establish the stated theorems.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

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